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Immunity, Infection and Forensics

Edexcel A A-Level Biology — Topic 6

6.1 Pathogens and Disease

Pathogen: A microorganism that causes disease. Pathogens include bacteria, viruses, fungi and protists. They cause disease either by producing toxins (exotoxins released by living cells, endotoxins from cell walls of Gram-negative bacteria) or by directly damaging host cells and tissues.

Disease vs Infection: An *infection* occurs when a pathogen enters the body and reproduces. A *disease* occurs when the infection causes recognisable symptoms due to cell damage or toxin production. Not all infections cause disease — some individuals can be asymptomatic carriers.

Types of Pathogen

Type	Cell Type	Key Features	Example Diseases	How They Cause Disease
Bacteria	Prokaryotic	Cell wall (peptidoglycan); reproduce by binary fission (as fast as every 20 min); some produce exotoxins/endotoxins; 0.2–10 µm	TB, cholera, MRSA, Salmonella	Exotoxins (e.g. cholera toxin opens Cl^- channels in gut) or endotoxins trigger fever and inflammation
Viruses	Acellular (non-living)	DNA or RNA core; protein capsid ± lipid envelope; obligate intracellular	HIV/AIDS, influenza,	Lyse host cells during replication (lytic cycle) or

		parasites; 20–300 nm; use host ribosomes to replicate	COVID-19, Ebola	integrate into host DNA (lysogenic cycle); trigger immune-mediated damage
Fungi	Eukaryotic	Chitin cell walls; heterotrophic (secrete enzymes externally); reproduce by spores; can be unicellular (yeast) or multicellular (moulds)	Athlete's foot, ringworm, aspergillosis	Secrete digestive enzymes onto host tissue; absorb nutrients; trigger inflammatory allergic responses
Protists	Eukaryotic	Single-celled; diverse morphology; often vector-transmitted; complex life cycles	Malaria (*Plasmodium*, vector: *Anopheles* mosquito)	*Plasmodium* invades red blood cells and liver cells; cells burst releasing merozoites; cycles of fever every 48–72 h

Modes of Transmission

- **Direct contact** — skin-to-skin, bodily fluids (STIs such as HIV, gonorrhoea; also Ebola through blood/bodily fluids)
- **Airborne (droplet/aerosol)** — coughing, sneezing produces droplets (>5 μm) or aerosol particles (<5 μm, remain suspended longer). Examples: TB, influenza, COVID-19
- **Waterborne** — contaminated water supply, often linked to poor sanitation. Examples: cholera (*Vibrio cholerae*), typhoid
- **Vector-borne** — transmitted by another organism (the vector). Malaria via *Anopheles* mosquito; Lyme disease via tick; plague via flea
- **Fomites** — contaminated surfaces or objects (door handles, medical equipment); norovirus and MRSA commonly spread this way
- **Foodborne** — contaminated or undercooked food. Salmonella in poultry, *E. coli* O157 in contaminated beef

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Koch's Postulates

In the 19th century, Robert Koch established criteria to prove that a specific microorganism causes a specific disease:

1. The microorganism must be found in abundance in all organisms suffering from the disease, but not in healthy organisms
2. The microorganism must be isolated from a diseased organism and grown in pure culture
3. The cultured microorganism should cause disease when introduced into a healthy organism
4. The microorganism must be re-isolated from the newly diseased organism and shown to be identical to the original

Limitations: Some pathogens cannot be cultured (viruses need host cells); some healthy individuals carry pathogens asymptomatically; ethical issues prevent deliberate infection of humans.

6.2 Non-Specific (Innate) Immune Response

The innate immune system provides immediate (within minutes to hours), non-specific defence against all pathogens. It does not confer immunological memory and responds the same way regardless of previous exposure. It is present from birth and is the first line of defence.

Physical and Chemical Barriers (First Line of Defence)

Barrier	Mechanism	Additional Detail
Skin	Physical barrier; outer layer of dead keratinised cells resistant to penetration	Low pH from sebum (~pH 5.5); antimicrobial peptides (defensins); microbiome outcompetes pathogens; intact skin is nearly impenetrable
Mucous membranes	Mucus traps pathogens; ciliated epithelium (mucociliary escalator) sweeps mucus towards throat	Lines respiratory tract, GI tract, urogenital tract; goblet cells secrete mucus; cilia beat 10–20 times per second
Stomach acid	HCl creates pH 1–2 environment; denatures pathogen proteins and	Pepsin (optimum pH 2) also destroys ingested pathogens; this is why oral

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	enzymes	vaccines must be protected from stomach acid
Lysozyme	Enzyme in tears, saliva, nasal secretions; hydrolyses glycosidic bonds in bacterial peptidoglycan cell walls	Causes lysis of Gram-positive bacteria particularly; part of innate chemical defence
Commensal bacteria	Normal flora (~100 trillion bacteria) outcompete pathogens for resources and attachment sites	Produce bacteriocins; lower local pH; why broad-spectrum antibiotics can cause secondary infections (e.g. *Clostridium difficile*)
Blood clotting	Platelets aggregate at wound site; fibrinogen → fibrin mesh traps blood cells, sealing the wound	Prevents entry of pathogens through broken skin

The Inflammatory Response (Second Line of Defence)

When tissue is damaged or infected, the following cascade occurs:

1. Damaged cells and **mast cells** release **histamine** and other chemical mediators (cytokines, prostaglandins)
2. Histamine causes **vasodilation** (arterioles widen) and **increased permeability** of capillary walls
3. More blood flows to the area → redness and heat. Increased permeability allows plasma and white blood cells (neutrophils, macrophages) to leave the blood and enter the tissue (**diapedesis**)
4. Phagocytes are attracted to the site by chemical signals (**chemotaxis**)
5. Swelling (oedema) and pain result from fluid accumulation and pressure on nerve endings

The four cardinal signs of inflammation: **redness** (rubor), **heat** (calor), **swelling** (tumor), **pain** (dolor).

> **Exam Tip:** Do not confuse inflammation with infection. Inflammation is the body's non-specific response to tissue damage (which could be caused by infection, but also by injury, burns, or allergens). It is a protective mechanism that recruits immune cells to the damaged area.

Phagocytosis

1. Phagocyte (neutrophil or macrophage) detects pathogen, attracted by chemicals released at the site of infection (**chemotaxis**). Pathogen may be recognised by its PAMPs (pathogen-associated molecular patterns) binding to PRRs (pattern recognition receptors) on the phagocyte
2. Phagocyte engulfs pathogen by endocytosis (membrane extends pseudopodia around the pathogen), forming a **phagosome**
3. Lysosomes fuse with the phagosome, forming a **phagolysosome**
4. Hydrolytic enzymes (lysozyme, proteases, nucleases) and reactive oxygen species digest and destroy the pathogen
5. **Neutrophils** die after phagocytosis (forming pus); **macrophages** survive and present pathogen antigens on their surface via MHC II proteins, becoming **antigen-presenting cells (APCs)** that activate the adaptive immune response

Opsonisation

Opsonins are molecules (such as antibodies or complement proteins) that coat the surface of a pathogen, making it more recognisable to phagocytes. The opsonin acts as a bridge between the pathogen and the phagocyte receptor, dramatically increasing the rate of phagocytosis. This links innate and adaptive immunity.

6.3 Specific (Adaptive) Immune Response

Antigen: A molecule (usually a protein or glycoprotein) on the surface of a cell that can trigger an immune response. Self-antigens on body cells are recognised as "self" and tolerated. Foreign antigens (on pathogens, transplanted tissue, toxins) trigger an immune attack. Antigens are typically large molecules with a specific 3D shape that a complementary receptor/antibody can bind to.

Self vs Non-Self: Every nucleated body cell displays **MHC (Major Histocompatibility Complex)** proteins on its surface, which are unique to each individual (except identical twins). The immune system is trained during development to tolerate self-MHC markers. Any cell lacking the correct MHC is attacked. This is why organ transplant rejection occurs and why immunosuppressant drugs are needed.

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T Lymphocytes (Cell-Mediated Response)

T cells are produced in the **bone marrow** but mature in the **thymus** (where self-reactive T cells are destroyed — central tolerance). Each T cell has a unique **T-cell receptor (TCR)** that recognises a specific antigen presented on an MHC protein by an APC.

Type	Function	Key Detail
T helper cells (T_H), CD4+	Release cytokines that activate B cells, T killer cells, and macrophages; stimulate clonal expansion	Central coordinators of the immune response. HIV specifically targets and destroys T_H cells (binds CD4 receptor), which is why AIDS causes immunodeficiency
T killer/cytotoxic cells (T_C), CD8+	Destroy infected body cells, cancer cells, and transplanted cells	Release **perforin** (creates pores in target cell membrane) and **granzymes** (enter through pores and trigger apoptosis — programmed cell death)
T regulatory cells (T_{reg})	Suppress immune response after infection is cleared; prevent autoimmunity	Maintain tolerance to self-antigens; secrete inhibitory cytokines (IL-10, TGF-beta). Dysfunction leads to autoimmune diseases
T memory cells	Remain in body for years or decades; rapid response on re-infection	Responsible for immunological memory; can persist for a lifetime; divide rapidly on re-exposure to same antigen

Exam Tip: The cell-mediated response is particularly important against: (1) intracellular pathogens (viruses, TB bacteria that live inside cells), (2) cancer cells displaying abnormal antigens, (3) transplant rejection, (4) fungal infections. Remember: T cells cannot detect free-floating pathogens — they need antigens presented on MHC proteins.

B Lymphocytes (Humoral Response)

B cells mature in the **bone marrow** and each produces a unique **antibody** (immunoglobulin) displayed on its surface as a B-cell receptor. The humoral response targets pathogens in body fluids (blood, lymph, tissue fluid).

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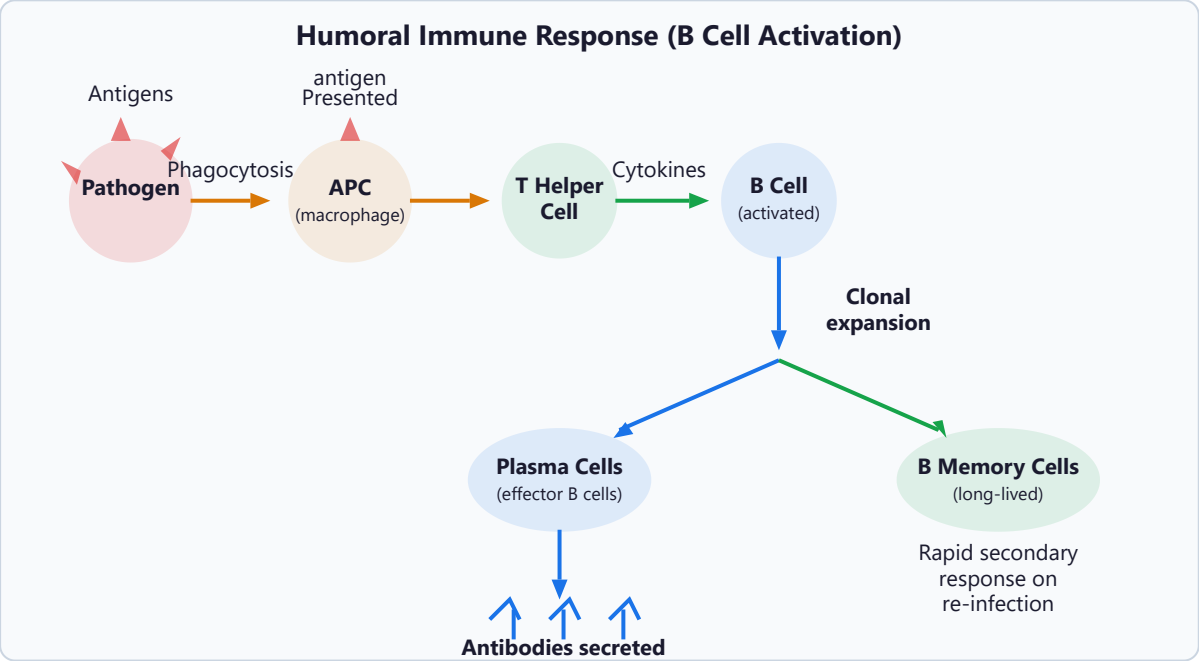
1. A B cell with a complementary antibody on its surface encounters and binds to a specific antigen (the B cell may also engulf and present the antigen on its MHC II)
2. A T helper cell with a complementary TCR binds to the antigen-MHC II complex on the B cell and releases **cytokines** (interleukins) that activate the B cell
3. The activated B cell undergoes **clonal expansion** (rapid mitotic division producing many identical clones)
4. Most clones differentiate into **plasma cells** (effector B cells) that secrete large quantities of specific antibodies (up to 2,000 molecules per second per cell) into the blood and lymph
5. Some clones become **B memory cells** that persist long-term and enable a faster, stronger secondary response

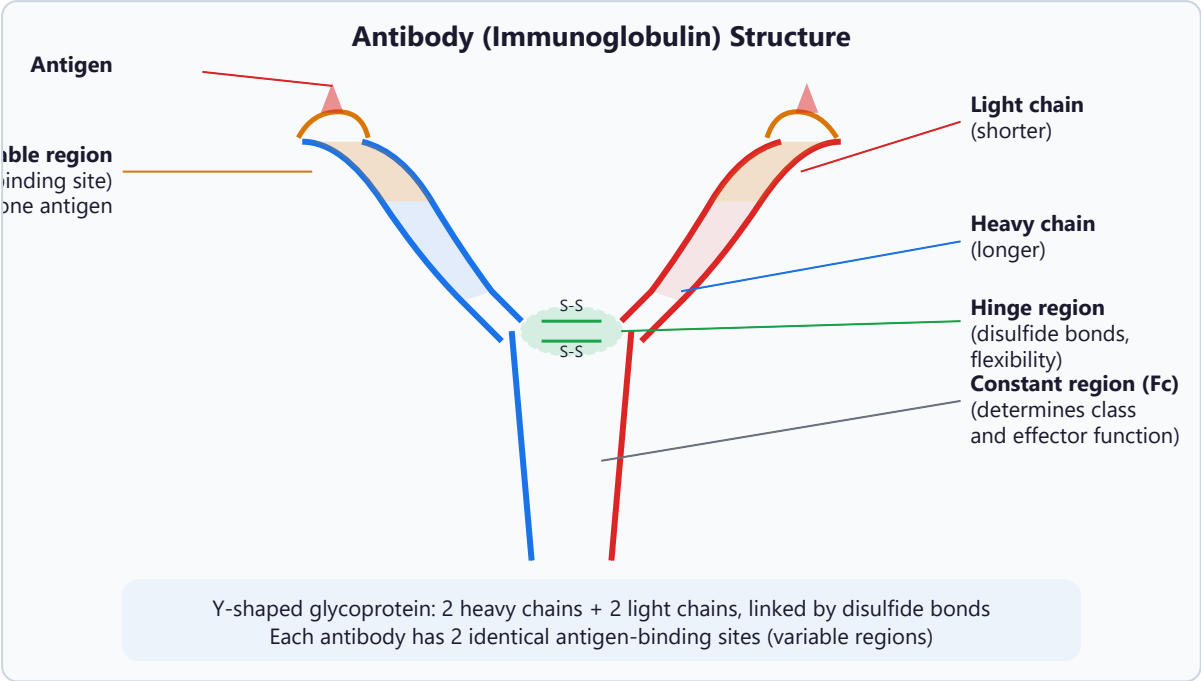


Antibody Structure and Function

Antibodies (immunoglobulins) are Y-shaped glycoproteins consisting of **two identical heavy chains** and **two identical light chains**, held together by **disulfide bonds**. The molecule has two distinct functional regions:

- **Variable region (Fab)** — at the tips of the Y; forms the **antigen-binding site**. The amino acid sequence varies enormously between antibodies, giving each a unique 3D shape complementary to one specific antigen. Each antibody has two identical binding sites
- **Constant region (Fc)** — the stem of the Y; determines the **class** (IgG, IgA, IgM, IgE, IgD) and **effector function** (e.g. ability to cross the placenta, activate complement, bind to mast cells)

- **Hinge region** — provides flexibility, allowing the two arms to move independently and bind antigens at different angles



How Antibodies Destroy Pathogens

Antibodies do not directly kill pathogens. Instead, they work by several mechanisms:

- **Agglutination** — antibodies (especially IgM, which is pentameric with 10 binding sites) cross-link pathogens, clumping them together into large complexes that are easier for phagocytes to engulf and that cannot enter host cells
- **Neutralisation** — antibodies bind to toxin molecules or to viral attachment proteins, blocking their ability to interact with host cell receptors. For example, antibodies against the influenza haemagglutinin protein prevent the virus from attaching to respiratory epithelial cells
- **Opsonisation** — antibodies coat the pathogen surface; the Fc region of bound antibodies is recognised by Fc receptors on phagocytes, greatly enhancing phagocytosis (up to 4,000-fold increase)
- **Complement activation** — the Fc region of bound antibodies triggers the classical complement cascade, a series of ~30 plasma proteins that form a **membrane attack complex (MAC)**, creating pores in the pathogen membrane and causing osmotic lysis

Antibody Classes

Class	Location	Key Function
IgG	Blood, tissue fluid	Most abundant (~75%); crosses placenta (passive immunity to fetus); activates complement; long-lived
IgA	Secretions (saliva, tears, breast milk, mucus)	Protects mucosal surfaces; main antibody in breast milk (passive immunity to newborn)
IgM	Blood	First antibody produced in primary response; pentamer with 10 binding sites — very effective at agglutination
IgE	Bound to mast cells in tissues	Triggers histamine release in allergic reactions; defence against parasites (helminths)
IgD	B cell surface	Functions as B cell receptor; role in B cell activation

Primary vs Secondary Immune Response

Feature	Primary Response	Secondary Response
Speed	Slow (7–14 days to peak)	Rapid (1–3 days to peak)
Antibody quantity	Lower peak concentration	Much higher concentration (100–1000x more)
Main antibody class	IgM initially, then IgG	Predominantly IgG
Cells involved	Naive B and T cells (clonal selection + expansion needed)	Memory B and T cells (already expanded, higher numbers)
Duration of antibody production	Short-lived (weeks)	Longer-lasting (months to years)
Affinity of antibodies	Lower affinity	Higher affinity (affinity maturation occurs)

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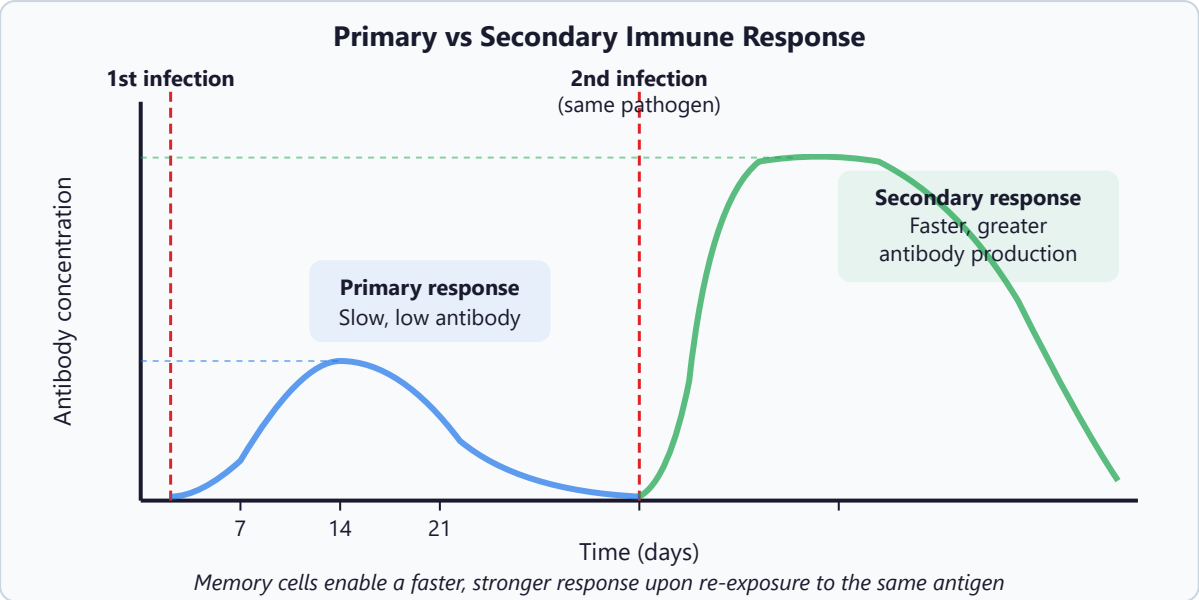

Exam Tip: The secondary response is faster and stronger because: (1) memory cells are present in much higher numbers than naive cells, (2) they have already undergone clonal selection so no delay in finding the right clone, (3) they have a lower activation threshold, (4) they can rapidly differentiate into effector cells without needing the full activation process. Questions often ask you to explain the shape of the antibody concentration graph — always refer to memory cells.

6.4 Vaccination and Immunity

Vaccination (immunisation): The deliberate administration of antigenic material (vaccine) to stimulate the adaptive immune response and the production of memory cells, providing long-term protection against future infection by the same pathogen, without causing the disease itself.

Types of Vaccine

Type	Description	Example	Pros/Cons
Live attenuated	Weakened form of pathogen that	MMR, oral polio, BCG (TB)	Strong, long-lasting immunity (mimics real infection); risk for

	replicates but does not cause disease		immunocompromised individuals; needs cold storage
Inactivated (killed)	Pathogen killed by heat/chemicals; cannot replicate	Influenza (injected), hepatitis A, rabies	Safer; weaker response; often needs booster doses
Subunit/conjugate	Purified antigen (protein or polysaccharide) from pathogen	Hepatitis B (HBsAg), HPV, meningococcal	Very safe; precise immune targeting; may need adjuvant
Toxoid	Inactivated toxin (formaldehyde-treated)	Tetanus, diphtheria	Targets the toxin rather than the organism itself
mRNA	mRNA encoding pathogen protein; host cells translate it and display the antigen	Pfizer/BioNTech and Moderna COVID-19 vaccines	Rapid development; does not contain live virus; needs ultra-cold storage (some formulations)

Adjuvants and Booster Doses

Adjuvants (e.g. aluminium salts) are substances added to vaccines to enhance the immune response by prolonging antigen exposure, attracting more immune cells to the injection site, or stimulating cytokine release. **Booster doses** are given to re-expose the immune system to the antigen, stimulating memory cells to proliferate again, maintaining high antibody levels and long-term protection.

Types of Immunity

Type	How Acquired	Memory Cells?	Duration	Example

Active natural	Infection with pathogen	Yes	Long-lasting (often lifelong)	Catching chickenpox
Active artificial	Vaccination	Yes	Long-lasting (may need boosters)	MMR vaccine
Passive natural	Antibodies across placenta (IgG) or in breast milk (IgA)	No	Temporary (weeks to months)	Newborn immunity
Passive artificial	Injection of purified antibodies	No	Temporary (weeks)	Antivenom for snakebite; rabies immunoglobulin

Exam Tip: A common exam question asks why passive immunity is temporary: the recipient's own immune system has not been activated, so no memory cells are produced. The injected antibodies are gradually broken down by proteases and not replaced. Active immunity is long-lasting because memory cells persist and can rapidly produce new antibodies on re-exposure.

Herd Immunity

When a sufficiently high proportion of a population is immune (through vaccination or prior infection), the pathogen cannot find enough susceptible hosts to maintain transmission chains. This indirectly protects unvaccinated individuals (those too young, immunocompromised, or with medical contraindications). The threshold varies by pathogen:

- **Measles:** ~95% (extremely infectious, R_0 = 12–18)
- **Polio:** ~80–85%
- **Influenza:** ~75–85%

Ethical Considerations of Vaccination

- Balancing individual rights (refusal of treatment) against public health (herd immunity protects vulnerable people)
- Rare adverse effects weighed against massive public health benefit

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- Equitable global distribution: many LIDCs lack access to vaccines that are routine in HICs
- Use of animal testing in vaccine development
- Mandatory vs voluntary vaccination programmes

6.5 Antibiotics and Resistance

Antibiotic: A substance produced by or derived from a microorganism that kills (**bactericidal**) or inhibits the growth of (**bacteriostatic**) bacteria. Antibiotics target specific bacterial structures or metabolic processes that are absent in human cells, which is why they can selectively harm bacteria without damaging the host. Antibiotics do **NOT** work against viruses because viruses lack their own metabolic machinery, cell walls, and ribosomes — they use host cell components instead.

Mechanisms of Antibiotic Action

Target	Mechanism	Example	Type
Cell wall synthesis	Prevents cross-linking of peptidoglycan; weakened wall leads to osmotic lysis	Penicillin, amoxicillin, vancomycin	Bactericidal
Protein synthesis (30S ribosome)	Binds to 30S subunit of 70S bacterial ribosome; causes misreading of mRNA or blocks tRNA binding	Tetracycline, streptomycin	Mostly bacteriostatic
Protein synthesis (50S ribosome)	Binds to 50S subunit; prevents translocation or peptide bond formation	Erythromycin, chloramphenicol	Bacteriostatic
DNA/RNA synthesis	Blocks DNA replication (DNA gyrase) or transcription (RNA polymerase)	Ciprofloxacin, rifampicin	Bactericidal

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Cell membrane	Disrupts membrane integrity; increases permeability leading to leakage of cell contents	Polymyxins	Bactericidal
Metabolic pathways	Competitive inhibitor of folic acid synthesis (needed for DNA synthesis)	Trimethoprim, sulfonamides	Bacteriostatic

Exam Tip: A common 6-mark question asks why antibiotics work against bacteria but not viruses. Key points: (1) bacteria have their own ribosomes (70S, different from human 80S) that antibiotics can target, (2) bacteria have peptidoglycan cell walls that antibiotics can disrupt, (3) viruses lack these structures — they use host cell machinery to replicate, (4) antiviral drugs exist but work by different mechanisms (e.g. inhibiting viral enzymes like reverse transcriptase).

Antibiotic Resistance

Antibiotic resistance is one of the greatest threats to global public health. The mechanism of resistance evolution follows Darwinian natural selection:

1. Within any bacterial population, there is **genetic variation** due to random mutations in DNA (and horizontal gene transfer). Some mutations may confer resistance — these exist *before* any antibiotic exposure
2. When an antibiotic is applied, it acts as a **selection pressure**: susceptible bacteria are killed or inhibited
3. Resistant bacteria survive and reproduce with reduced competition for resources (**survival of the fittest**)
4. The **resistance allele frequency increases** in the population over subsequent generations (bacteria reproduce rapidly — generation time can be 20 minutes)
5. **Horizontal gene transfer** spreads resistance: conjugation (plasmid transfer between bacteria via pili), transduction (bacteriophage carries gene between bacteria), transformation (uptake of free DNA from environment). Resistance can spread between different species

Mechanisms of Resistance

- **Altered target site** — mutation changes the protein the antibiotic normally binds to (e.g. altered penicillin-binding proteins in MRSA)
- **Efflux pumps** — membrane proteins actively pump the antibiotic out of the bacterial cell before it can act
- **Enzymatic destruction** — bacteria produce enzymes that break down the antibiotic (e.g. beta-lactamase destroys penicillin's beta-lactam ring)
- **Reduced permeability** — changes to outer membrane porins reduce antibiotic entry
- **Alternative metabolic pathway** — bacteria use a different enzyme that is not inhibited by the antibiotic

Combating Resistance

- Complete full courses of antibiotics (prevents partially resistant bacteria surviving)
- Do not use antibiotics for viral infections
- Restrict use of antibiotics in agriculture and animal farming
- Develop new antibiotics and alternative therapies (e.g. bacteriophage therapy, antimicrobial peptides)
- Improved diagnostic tests to identify specific pathogens and targeted treatment
- Antibiotic stewardship programmes in hospitals

> **Exam Tip:** Always emphasise that bacteria do NOT "become resistant" in response to antibiotics. Resistance arises from random pre-existing mutations; antibiotics simply act as a selection pressure that favours resistant individuals. This is evolution by natural selection — not Lamarckian inheritance. Examiners will penalise teleological language such as "bacteria develop resistance to survive."

6.6 Forensic Biology

DNA Profiling (Genetic Fingerprinting)

DNA profiling: A technique that analyses **short tandem repeats (STRs)** — short non-coding repetitive DNA sequences (2–6 base pairs repeated 5–50 times) that vary in the number of repeats between individuals — to produce a unique pattern (profile) for identification. STRs are inherited in a Mendelian fashion (co-dominant). The UK national DNA database analyses 16+ STR loci for forensic identification.

Steps in Modern DNA Profiling

1. **DNA extraction** from biological sample (blood, saliva, hair follicle with root, semen, skin cells). Even tiny amounts of DNA can be used thanks to PCR amplification
2. **PCR amplification** of specific STR loci. Primers flanking each STR region are used. Fluorescently labelled primers allow detection. Typically 16+ loci are amplified simultaneously (multiplex PCR)
3. **Capillary electrophoresis** separates the amplified fragments by size. Smaller fragments migrate faster through the polymer-filled capillary under an electric field
4. **Fluorescent detection:** A laser detects the fluorescent labels on fragments as they pass a detector, producing an electropherogram (peaks at specific sizes)
5. **Analysis:** The pattern of peaks (alleles at each locus) is compared between samples. A match at all loci tested provides extremely high confidence of identity

Note: The older technique used restriction enzymes to cut DNA at recognition sequences, gel electrophoresis to separate fragments, Southern blotting to transfer to a membrane, and radioactive/fluorescent probes to visualise VNTRs. Modern profiling using PCR and capillary electrophoresis is faster, more sensitive, and requires smaller samples.

Applications of DNA Profiling

- **Criminal investigations** — matching suspects to crime scene evidence
- **Paternity/maternity testing** — child must share alleles with each parent at every locus
- **Identification of disaster victims** — comparison with relatives or pre-existing DNA records
- **Immigration disputes** — verifying family relationships
- **Evolutionary relationships** — comparing STR profiles between species or populations
- **Exonerating wrongly convicted individuals** — the Innocence Project has freed over 375 people using DNA evidence

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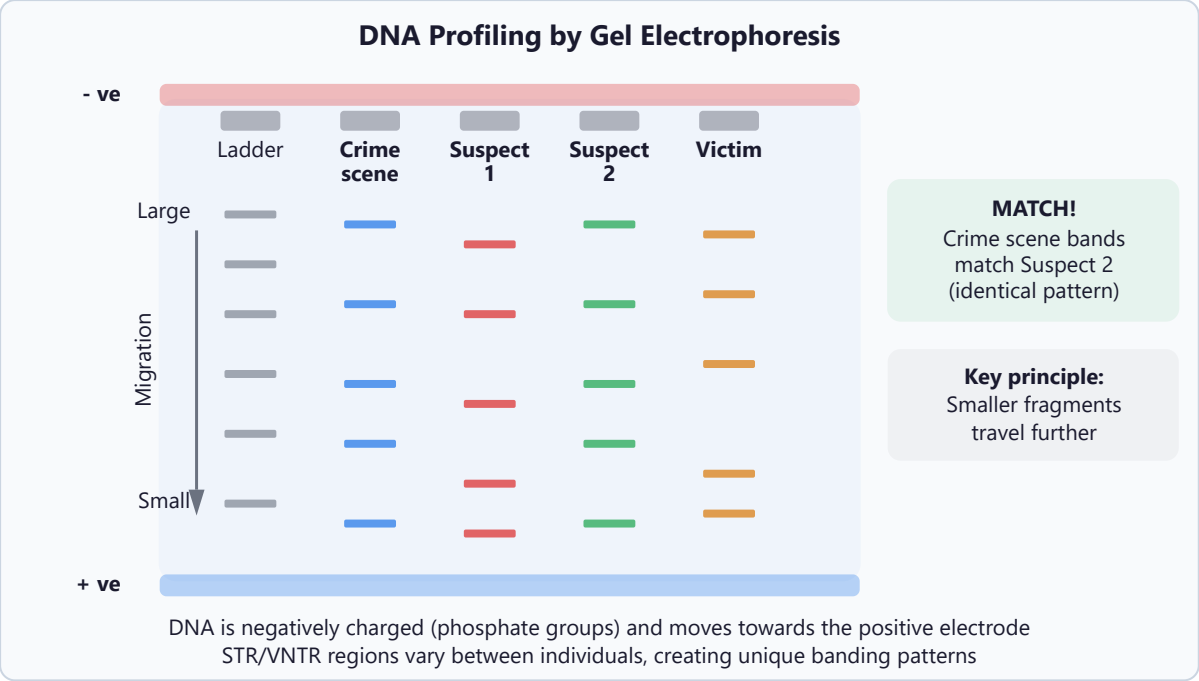
Gel Electrophoresis



In gel electrophoresis, DNA fragments are placed in wells of an **agarose gel** submerged in buffer solution and an electric current is applied. DNA molecules carry a **negative charge** (due to phosphate groups in the sugar-phosphate backbone), so they migrate towards the **positive electrode (anode)**. The gel acts as a molecular sieve: **smaller fragments move faster and further** because they encounter less resistance passing through the gel pores. Fragments are visualised with UV light (if stained with ethidium bromide or SYBR Green) or by fluorescent labelling.

> **Exam Tip:** In DNA profiling questions: (1) identical twins have the same profile (same DNA), (2) children share approximately half their bands with each parent (one allele from each parent at each locus), (3) the probability of two unrelated individuals having the same profile is extremely low — typically less than 1 in a billion when 16+ STR loci are tested, (4) DNA profiling does not prove guilt on its own — it places the individual at the scene but does not explain how the DNA got there (could be contamination or legitimate presence).

Estimating Time of Death

Forensic scientists use multiple methods in combination to estimate the time since death (post-mortem interval), as no single method is fully reliable:

Method	Principle	Timescale	Limitations
Body temperature (algor mortis)	Body cools at approximately 1.5°C/hour (Newton's law of cooling) until reaching ambient temperature	First 24 h	Affected by body mass, clothing, ambient temperature, air movement, body fat, fever at death
Rigor mortis	Muscles stiffen due to ATP depletion and actin-myosin cross-bridge formation. Onset 2–4 h; fully rigid by 12 h; begins to resolve at 24–36 h; fully resolved by 48–60 h	2–60 h	Affected by temperature (faster in heat), physical exertion before death, body mass
Livor mortis (hypostasis)	Blood pools in lowest parts of body due to gravity; initially blanchable (pressing turns skin pale), becomes fixed after 8–12 h	0.5–12 h	Skin colour may obscure; position changes before fixing affect distribution
Degree of decomposition	Predictable stages: fresh (0–2 days) → bloat (2–6 days) → active decay (6–10 days) → advanced decay (10–25 days) → dry/skeletal remains (>25 days)	Days to months	Hugely affected by temperature, humidity, oxygen availability, burial, clothing
Forensic entomology	Specific insects colonise corpses in a predictable succession. Blowflies arrive within minutes. Larval (maggot) developmental stages and species present indicate time elapsed	Hours to months	Temperature-dependent development; geographical variation in species; access may be restricted
Stomach contents	Food type and degree of digestion indicate time since last meal. Stomach empties approximately 4–6 h after eating	Hours	Must know time of last meal; variation between individuals and food

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			types; drugs/illness affect gastric emptying
Vitreous humour potassium	Potassium concentration in the eye's vitreous humour increases predictably after death as cell membranes break down	24–100 h	Affected by temperature; requires laboratory analysis

Exam Tip: When answering questions on time of death estimation, always emphasise that (1) multiple methods should be used in combination for accuracy, (2) each method has limitations and confounding variables, (3) environmental conditions (especially temperature) are the biggest source of error across all methods. Examiners reward answers that discuss limitations as well as principles.

Decomposition and Succession

Decomposition involves both **autolysis** (self-digestion by the body's own enzymes, particularly from lysosomes) and **putrefaction** (bacterial breakdown, especially by anaerobic gut bacteria). The progression of insect colonisation follows a predictable ecological succession:

1. **Blowflies** (*Calliphora*, *Lucilia*) arrive within minutes to hours; lay eggs in natural orifices and wounds
2. **Flesh flies** and other dipterans arrive during bloat stage
3. **Beetles** (e.g. *Dermestes*, hide beetles) arrive during later decay stages when tissue dries
4. **Moths and mites** colonise during dry/skeletal stage

By identifying the species present and measuring the developmental stage of larvae (instar stages), entomologists can calculate a **minimum post-mortem interval** (minimum PMI) using accumulated degree hours or days (ADH/ADD) adjusted for ambient temperature.